



The Volcker Rule and corporate bond market making in times of stress[☆]



Jack Bao^d, Maureen O'Hara^{b,c,*}, Xing (Alex) Zhou^a

^a Federal Reserve Board, 20th St and Constitution Ave NW, Washington, DC 20551, USA

^b Samuel Curtis Johnson Graduate School of Management, Cornell University, 447 Sage Hall, Ithaca, NY 14853, USA

^c University of Technology Sydney, Sydney, Australia

^d University of Delaware, Lerner College of Business and Economics, Department of Finance, 42 Amstel Avenue, Newark, DE 19716, USA

ARTICLE INFO

Article history:

Received 20 December 2016

Revised 7 June 2017

Accepted 30 June 2017

Available online 15 June 2018

JEL classification:

G14

G21

G23

G24

G28

Keywords:

Volcker Rule

Corporate bond illiquidity

Regulation

Capital commitment

Dealer inventory

Market making

ABSTRACT

Focusing on downgrades as stress events that drive the selling of corporate bonds, we show that the illiquidity of stressed bonds has increased after the Volcker Rule. Dealers regulated by the rule have curtailed their market-making activities and non-Volcker-affected dealers have not offset the decreased activities of Volcker-affected dealers. Furthermore, even Volcker-affected dealers that are not constrained by Basel III and Comprehensive Capital Analysis and Review regulations change their behavior, inconsistent with the effects being driven by these other regulations. Because Volcker-affected dealers have been the main liquidity providers, bonds have become less liquid during times of stress due to the Volcker Rule.

© 2018 Elsevier B.V. All rights reserved.

[☆] We thank an anonymous referee, Manuel Adelino, Michael Barr, Sergey Chernenko, Jens Christensen (San Francisco Fed Financial Structure and Regulation discussant), Dan Covitz, Alie Diagne, Darrell Duffie, Amy Edwards (Conference on Financial Market Regulation at the SEC discussant), Peter Feldhutter, Josh Gallin, Jeff Gordon, Song Han, Bob Hockett, Kewei Hou, Howell Jackson, Adam Kolasinski, Jongsub Lee, Nellie Liang, Kleopatra Nikolaou, Michael Palumbo, Andreas Park (Conference on Financial Economics and Accounting (CFEA) discussant), Mitchell Petersen, Doriana Ruffino, Mehrdad Samadi (Financial Intermediation Research Society (FIRS) discussant), Krista Schwarz (Wharton Conference on Financial Regulation and Rule of Law discussant), Meg Tahyar, Haoxiang Zhu (Notre Dame Conference on Current Topics in Financial Regulation discussant), conference participants at the 2016 CFEA, 2016 Notre Dame Conference on Current Topics in Financial Regulation, 2016 San Francisco Fed Financial Structure and Regulation Conference, Fourth Annual Conference on Financial Market Regulation at the SEC, 2017 FIRS Conference, 2017

"It may be also that, even if liquidity is adequate in normal conditions, it has become more fragile, or prone to disappearing under stress."

— Governor Jerome H. Powell,

Board of Governors of the Federal Reserve System

Wharton Conference on Financial Regulation and the Rule of Law, and seminar participants at the Federal Reserve Board, MIT Sloan School of Management, the Office of Financial Research, the SEC, and Yale School of Management for helpful comments and discussions. The views expressed herein are ours and not necessarily of the Federal Reserve Board of Governors or its staff. All errors are our own.

* Corresponding author.

E-mail address: mo19@cornell.edu (M. O'Hara).

1. Introduction

Among the many regulatory changes following the 2007–2009 financial crisis, few are more controversial than the Volcker Rule. Enacted as part of the 2010 Dodd–Frank Wall Street Reform and Consumer Protection Act, the Volcker Rule was intended to limit bank risk taking by restricting or prohibiting certain speculative activities. Critics (e.g., [Duffie, 2012](#)) contended that an unintended consequence of the rule could be diminished bond market liquidity, resulting from a reduction in banks' market-making activities. Advocates of the rule disagreed, arguing that non-Volcker-affected dealers could compensate for any market-making reductions, leaving liquidity essentially unchanged. Settling this debate has taken on new urgency given the current discussions surrounding repeal or revision of the Volcker Rule by the Trump administration.¹

In this paper, we provide an empirical analysis of the impact of the Volcker Rule on liquidity for corporate bonds, particularly during stress events. We focus on determining if bond market liquidity has changed post-Volcker in such stress periods and, if so, why. Practitioners and policy makers alike have noted that illiquidity in times of stress is the more relevant metric for gauging market stability and performance, as that is when liquidity is needed most.² Motivated by [Ambrose et al. \(2008\)](#) and [Ellul et al. \(2011\)](#), who find evidence of forced selling of downgraded bonds induced by regulatory constraints imposed on insurance companies, we use downgrades of corporate bonds to junk status as stress events in which liquidity is demanded by clients.³ Concentrating on regulation-induced sales has the added advantage of plausibly preventing investors from optimally timing their trades, thereby providing a more reliable estimate of the liquidity conditions that investors face in times of stress. Furthermore, the reporting metrics imposed by the Volcker Rule have direct implications about the willingness of affected dealers to take on large inventories of any individual bonds that would be hard to turn over, exactly the liquidity provision needed around regulation-induced fire sales.

Our focus is on a difference-in-differences test comparing the illiquidity of downgraded corporate bonds with a baseline control group both before and after the Volcker Rule was implemented. The first difference is the difference in price impact between a set of bonds recently downgraded to speculative grade from investment grade and a set of BB bonds used to control for the general

level of illiquidity. The second difference is between the post-Volcker difference and the pre-Volcker difference. Our main result is that bond liquidity deterioration around rating downgrades has worsened following the implementation of the Volcker Rule. We find these adverse effects whether we benchmark to the precrisis period or to the period just before the Volcker Rule was enacted.

We then turn to understanding why these effects have arisen by investigating Volcker Rule – induced changes in dealer behavior, with a focus on any differential effects on Volcker-affected versus non-Volcker-affected dealers. Because the Volcker Rule applies to only banks with access to government backstops (such as deposit insurance or Federal Reserve borrowing), other dealers without such access are unlikely to reduce their market making in response to the Volcker Rule. In principle, they could even step in to provide liquidity in cases in which the lines between permissible market making and prohibited proprietary trading are blurred. Our second major result is that liquidity deterioration in post-Volcker stress periods featured less liquidity provision by Volcker-affected dealers, with only weak evidence of increased liquidity provision by non-Volcker-affected dealers.

We show that liquidity provision by Volcker-affected dealers dropped during post-Volcker stress times and by non-Volcker-affected dealers has not increased enough to compensate for this decline. In the post-Volcker period, the relative share of dealer–customer trades taken by non-Volcker dealers increased. Dealers affected by the Volcker Rule see an economically and statistically significant increase in agency trades, or trades in which the dealer pre-arranged an offsetting trade so as not to have inventory risk. They have also committed significantly less capital in market making. For non-Volcker dealers, no such effects are evident on both agency trades and capital commitment in the post-Volcker period. Combined with our results on the increased illiquidity during the post-Volcker period, these results demonstrate that while non-Volcker dealers hypothetically could step in and provide additional liquidity in the long run, concerns about the unintended effects of the Volcker Rule were well founded as such change has not yet happened.⁴ At least during our sample period, no evidence exists of sufficient additional liquidity provision by non-Volcker-affected dealers to offset the decreased liquidity in bank bond market trading.

A natural concern is whether our results are related to the Volcker Rule or instead arise from other important regulations on dealer bond market behavior, such as Basel III and Comprehensive Capital Analysis and Review (CCAR) stress testing. To disentangle these effects, we focus on the implementation period of the Volcker Rule as compared with the period just before implementation. We also split dealers by their exposure to Basel III. Though most banks' capital ratios are significantly above Basel III minimums, increased Basel III capital requirements along with CCAR constraints could induce some banks to reduce their market-making activities. These CCAR constraints

¹ See, for example, [Tracy and Carney \(2016\)](#), [DiPietro \(2016\)](#) and [New York Times \(2016\)](#).

² See recent comments in [Deutsche Bank Research \(2016\)](#) and testimony in [Powell \(2016\)](#) that even if liquidity is high in normal conditions, it can become more troublesome in periods of stress.

³ Though our paper focuses on firm-level stress events, it is well-known that liquidity deteriorates significantly in macroeconomic stress events. See, for example, [Bao et al. \(2011\)](#), [Dick-Nielsen et al. \(2012\)](#), and [Friewald et al. \(2012\)](#) for evidence of liquidity deterioration during the recent financial crisis. See also [Eom et al. \(2004\)](#) and [Huang and Huang \(2012\)](#) for a broader discussion of corporate bond illiquidity and bond pricing. [Acharya and Richardson \(2009\)](#) argue that insurance companies can pose systemic risks through their holding of corporate bonds. See also [Huang et al. \(2016\)](#) for a discussion of insurer corporate bond holdings.

⁴ See [Federal Register \(2014\)](#) on the Volcker Rule for details of comment letters. Liquidity deterioration was particularly severe during the height of the financial crisis.

require dealers to meet minimum capital requirements in stress scenarios. To address this concern, we split Volcker-affected dealers into those that are CCAR-constrained and those that are not. We find that capital commitment has decreased significantly for dealers that neither have failed CCAR tests nor been given a conditional pass. Thus, our results are unlikely to be driven by banks adjusting their business to remedy failed CCAR tests.

Our paper is most closely related to several recent studies on regulation and liquidity, all of which focus on the general regulatory environment following the global financial crisis. Studying general trends in corporate bond market liquidity, Trebbi and Xiao (2015) argue that liquidity has not deteriorated following post-crisis regulations. Bessembinder et al. (2016) similarly find no deterioration in liquidity following post-crisis regulations.⁵ Their results suggest that despite some dealers providing less liquidity, little evidence shows that regulation leads to an increased cost of trading in normal times. By studying liquidity metrics around rating downgrades, a recent paper by Anderson and Stulz (2017) does not find evidence of lower liquidity during the post-crisis period. Dick-Nielsen and Rossi (2016) study liquidity provision around index exclusion events, finding that liquidity has deteriorated post-crisis. All four papers provide evidence of how liquidity has changed following the passage of crisis-era reform rules. In contrast, we focus on liquidity in stress periods and on isolating the specific effects arising from implementation of the Volcker Rule. Our main results relate to comparing the post-Volcker implementation period with the period just before Volcker implementation, whereas both Bessembinder et al. (2016) and Dick-Nielsen and Rossi (2016) focus on the years prior to Volcker Rule implementation.⁶ The Trebbi and Xiao (2015) sample extends to the end of 2014, but it does not focus on the Volcker Rule implementation period. Because the Anderson and Stulz (2017) sample also ends in 2014, its primary results are about the post-Dodd-Frank period and are consistent with our finding that liquidity is improving in this period. Furthermore, our use of the regulatory version of the Trade Reporting and Compliance Engine (TRACE), with dealer identities, allows us not only to split dealers by those that are directly affected by the Volcker Rule and those that are not, but also to identify which dealers were potentially affected by other regulations such as Basel III and CCAR.⁷

The evidence in our study suggests that there are significant costs to the proprietary trading ban in the Volcker Rule, and we consider how the Volcker Rule could be improved in light of our results. However, we do not address two issues. First, we do not do any welfare analysis to assess whether the Volcker Rule is overall net positive or net negative for financial markets and the economy.⁸ A potential benefit of the Volcker Rule is the ban on risky trades by institutions that could eventually seek government support if their risky trades lead to significant losses. Such an analysis requires modeling the trade-off between the social cost to the loss of liquidity in corporate bond markets and the societal benefit of safer banks, which is beyond the scope of our study. Second, we do not speak directly to what would happen in a financial crisis with the Volcker Rule. In a market crisis, dealers can react by withdrawing liquidity from all bond issues, not merely from the stressed issues considered here. Because such tail events happen rarely, the short time since the implementation of the Volcker Rule precludes any sensible analysis. However, the fact that dealers have become less willing to provide liquidity for single firm-specific negative shocks even in normal macroeconomic conditions is worrisome. In more adverse conditions, when bonds issued by multiple firms need liquidity support, we think it unlikely that the collective dealer community will be willing or able to do so.

The rest of the paper is organized as follows. In Section 2, we discuss the Volcker Rule and its potential impact on market making in the corporate bond market. In Section 3, we describe our data sources and variable construction. In Section 4, we examine changes in liquidity around stress events and provide a variety of robustness tests. In Section 5, we examine how the behavior of Volcker-affected and non-Volcker-affected dealers change with the implementation of the Volcker Rule. We also discuss Basel III and CCAR regulations. Section 6 concludes with a discussion of potential changes to the Volcker Rule.

2. Potential impact of the Volcker Rule on the corporate bond market

As part of the Dodd-Frank Act, enacted on July 21, 2010, Section 13 (the Volcker Rule) was added to the Bank Holding Company Act of 1956. Section 13 generally prohibits banking entities from engaging in proprietary trading or having ownership or relations with hedge funds and private equity funds. Implementation of Section 13 was not immediate and followed a laborious process. On January 18, 2011, the Financial Stability Oversight Council (2011) offered recommendations for implementing Section 13. The US Department of the Treasury, Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation (FDIC), and Securities and Exchange Commission

⁵ A recent update of Bessembinder et al. (2016) has added new data and tests of the 2014–2016 period, finding results that, like ours, show that dealer liquidity provision and bond liquidity have declined after Volcker Rule implementation.

⁶ Both papers discuss their results as being related to an anticipation of new regulations. Our results look at the implementation of the Volcker Rule. While changes in behavior due to an anticipation of regulations are possible, we believe that expecting large-scale changes in behavior around implementation is reasonable.

⁷ A few other recent papers focus on dealer behavior. Schultz (2017) finds, like us, that the Volcker Rule has affected the willingness of dealers to take on inventory. Adrian et al. (2017) find that more levered institutions tend to provide less liquidity after the crisis, leading to lower liquidity. Choi and Huh (2017) argue that a shift has occurred away from dealer liquidity provision to customer liquidity provision. Goldstein and Hotchkiss (2017) provide a broad study of dealer liquidity provision.

⁸ Costs do exist for not having regulation. For example, Chernobai et al. (2016) show that operational risk events increased during the gradual deregulation of bank holding companies from 1996 to 1999. On the other hand, Frydman and Hilt (2017) show that the Clayton Antitrust Act of 1914, which prevented some bankers from serving on the boards of railroads, had a deleterious effect on these railroads. We are, however, unaware of any studies quantitatively measuring the costs of allowing banks to participate in proprietary trading.

(SEC) worked with the Commodities Futures Trading Commission (CFTC) in formulating a proposal before releasing a version for comments in the *Federal Register* in November 2011 ([Federal Register, 2011](#)). In December 2013, final regulations were issued, and final regulations with details of market participants' comments were released in the *Federal Register* on January 31, 2014 ([Federal Register, 2014](#)). On April 1, 2014, the Volcker Rule became effective, requiring that banks with at least \$50 billion in trading assets report some quantitative metrics starting in July 2014. By July 21, 2015, large banks were required to be fully compliant with the Volcker Rule. During the conformance period, banks had to make good faith efforts to conform to the new rules.⁹ Hence, at least some effects of the Volcker Rule are expected to become evident starting in April 2014.

Other research (e.g., [Bessembinder et al., 2016](#); [Dick-Nielsen and Rossi, 2016](#)) has argued that anticipation of new regulation implementation could lead to earlier changes in dealer behavior. While we find little evidence of a significant anticipation effect right before the rule implementation, our tests do not necessarily preclude the possibility of some changes in dealer behavior prior to rule implementation. Our tests are designed to gauge the additional impact of Volcker Rule implementation, mainly benchmarking to the period just before implementation.

The intent of the Volcker Rule is to prohibit banking entities with access to the discount window at the Federal Reserve or to FDIC insurance from engaging in risky proprietary trading. It is important to keep in mind that not all financial firms are covered. For example, an [Oliver Wyman and SIFMA \(2011\)](#) study lists Cantor Fitzgerald & Co., Daiwa Capital Markets, Jefferies & Co., and Nomura as explicitly not covered. Also, not all trading activities are precluded. Recognizing that some activities are necessary for the market to function normally, the Volcker Rule includes an explicit set of permitted activities. The most relevant one for this paper, and arguably the most controversial, is a provision that permits market making. Essentially, affected dealers can trade securities to facilitate client-driven transactions, but they cannot transact in a way intended to make profits based on the price appreciation of securities.

A major difficulty in implementing the market-making exception is distinguishing allowed market making from prohibited proprietary trading. The [Financial Stability Oversight Council \(2011\)](#) proposed a number of principles to distinguish between the two.¹⁰ Among these are that market making should have rapid inventory turnover with the vast majority of profits from bid-ask spreads instead of from price appreciation. Proprietary trading is likely to have more modest turnover with significant profits from price appreciation. The final law requires establishment of an internal compliance program and the reporting of seven sets of daily metrics: (1) risk and position limits and usage, (2) risk factor sensitivities, (3) value at risk (VaR) and stress VaR, (4) comprehensive profit and loss, (5) inventory

turnover, (6) inventory aging, and (7) customer facing trade ratio. Though banks generally do not have to report these metrics at a trade or bond level, some metrics are required to be reported at the sector level or for all new positions. Regardless, all large positions and trades have the potential to affect these metrics, which are used by regulators to provide an initial indication of whether a bank is likely to be proprietary trading.

An important consequence of these reporting metrics is that they disincentivize dealers from taking large order imbalances that cannot be easily unwound, as these order imbalances have the potential effect of distorting many of these metrics. For example, banks are required to report factor exposures at a granular enough level to account for specific credit sectors and market segments. Inventory turnover requires the reporting of 30-, 60-, and 90-day turnover. Inventory aging metrics require the reporting of the value of securities held at different horizons. Taking on large inventories of bonds that are difficult to sell distorts these metrics in a way that could raise concerns for regulators and bring additional scrutiny. Though some have conjectured that trading to absorb sales of bonds that customers are unloading would be recognized as market making (e.g., [Anderson and Stulz, 2017](#)), to date, no such exception allows for omission from the reporting metrics. Furthermore, the Volcker Rule contains backstop provisions that prevent even market-making activities if they, among other things, subject a banking entity to exposure to high-risk assets or trading strategies ([Barr et al., 2016](#)).

Critics of the Volcker Rule note many gray areas in the rule and further argue that ambiguity in how the rule would be enforced was likely to be detrimental to market liquidity. Furthermore, though the intent of market making and proprietary trading can be different, observationally, they are difficult to distinguish. In fact, some argue that proprietary trading could be deemed risky market making. [Duffie \(2012\)](#) and other commenters suggest that the Volcker Rule could be particularly problematic in illiquid markets such as corporate bond markets. Duffie notes that whereas the average half-life of order imbalance in equities is three days, for investment-grade corporate bonds it is roughly two weeks. Thus, metrics based on measures such as inventory aging and inventory turnover could be particularly problematic for market making in corporate bonds. Furthermore, dealers who fear violating the Volcker Rule could be unable to properly manage inventory. One of the guidelines for the Volcker Rule is meeting near-term customer demand. But absent perfect predictions about future customer demand, market makers could find it difficult to pre-position in advance of variation in customer trading.

The final rule also presents complications for fulfilling customer demand because of the required internal compliance metrics. [Federal Register \(2014, p. 5612\)](#) notes that trades exceeding internal limits “should not be permitted simply because it responds to customer demand.” Instead, a banking entity is required to have escalation procedures that include “demonstrable analysis and approval.” Such regulations mean that market makers will find responding to large sells in the market particularly difficult.

One initial proposal dropped from the final rule was a requirement for detailed revenue attribution. This included

⁹ See [Federal Reserve Board \(2016\)](#).

¹⁰ The original proposal by Volcker was based on the idea of an “I know it when I see it” test, but the rule has much more specific prescriptions ([Barr et al., 2016](#)).

identifying revenue attributable to the bid-ask spread as opposed to price appreciation. While the final rule no longer has such a requirement, it does have a profit and loss attribution requirement that focuses on revenue generation patterns. Abnormal patterns could raise a red flag and lead to further review. Given the illiquid nature and infrequent trading patterns in corporate bonds, this could cause issues for market makers, particularly when a significant subset of its bonds has a severe order imbalance.

In April 2017, Deutsche Bank became the first bank fined for Volcker Rule violations, incurring an almost \$157 million total fine (Hamilton, 2017). Deutsche Bank was fined for lax oversight of traders and an insufficient compliance program to ensure that their business did not violate the Volcker Rule. Even prior to this first regulatory action, anecdotal evidence suggests that regulated banks were concerned about potentially violating the Volcker Rule. In late 2016, Goldman Sachs compiled documents in defense of a trader who had notoriously made more than \$100 million trading junk bonds earlier in the year (Campbell and Natarajan, 2016). Thus, both regulator actions and bank actions in response to potential regulator actions suggest that the Volcker Rule poses a real challenge to how banks trade corporate bonds and potentially distorts the behavior of banks.

In summary, Volcker Rule requirements have the potential to impact the behavior of dealers covered by the rule and lead to less liquid markets. Ambiguity as to what is legal market making and what is prohibited proprietary trading can exacerbate the problem by pushing dealers toward more conservative trading strategies. New rules favoring customer-facing trades can discourage dealers from using the inter-dealer market, and inventory-based metrics can lead dealers to reduce their inventory exposure. Perhaps most pertinent to our study, the requirement that dealers set internal limits could result in dealers being unable to respond to increased customer demands during times of stress.

3. Data and sample description

To examine how the Volcker Rule has affected corporate bond liquidity in stress periods, we focus on bond trading when a bond is downgraded from investment grade to speculative grade. Insurance companies, the dominant investors in the corporate bond market, face regulatory restrictions when investing in the corporate bond markets. The National Association of Insurance Commissioners (NAIC) classifies corporate bonds into six risk categories (NAIC1–NAIC6) based on their credit ratings and requires insurance companies to maintain a higher level of capital when investing in bonds in a higher risk category.¹¹ In addition, insurance companies are usually required to invest no more than 20% of their assets in bonds below NAIC2, i.e., speculative-grade bonds. Ellul et al. (2011) find that rating downgrades to speculative

grade can trigger fire sales in the bond market because greater capital requirements and other regulatory constraints prompt widespread divestment by insurance companies. Such regulation-induced fire sales generate high demand for liquidity and can cause substantial price pressure in the absence of adequate liquidity provision.

We obtain the rating history file from the Mergent Fixed Income Securities Database (FISD) for the period from January 2006 to March 2016.¹² This data file provides the announcement date of rating actions by the three largest rating agencies: Standard & Poor's (S&P), Moody's, and Fitch. We focus on fixed coupon corporate bonds with semi annual coupon payments, \$1000 par amount, and fixed maturity. These bonds are issued in US dollars by US firms in the following three broad FISD industry groups: industrial, finance, and utility. We exclude from our sample the following bonds: convertible or puttable bonds, private placements, asset-backed issues, and issues that are part of a unit deal. Because rating agencies differ with respect to the timing of rating actions, we follow (Ellul et al., 2011) and define the rating change event as the date of downgrade from investment grade to speculative grade announced by the first acting rating agency.

We then extract data from the Financial Industry Regulatory Authority (FINRA) TRACE database on corporate bond transactions during the one month following each rating downgrade. Our choice of a one-month window follows from analyzing trading by insurance companies around rating downgrades. Using NAIC data on bond transactions by insurance companies, we find that volume spikes right after downgrade by the first acting rating agency (see Fig. 1). It becomes volatile but remains elevated on average for roughly a month before it peters out afterward. Because the median number of days between downgrade by the first and the second rating agency is 13 in our sample, our one-month window captures increased demand for liquidity after both rating actions.¹³ The window is also consistent with the price impact pattern as shown in Ellul et al. (2011).¹⁴ Overall, this suggests that a shorter window can fail to capture the full effect of demand for liquidity by insurers and a window longer than one month potentially blends in a period in which liquidity is demanded with one in which it is not.

TRACE data provide detailed information on all secondary market transactions in the downgraded bond, including bond CUSIP, trade execution date and time, trade price and quantity, a buy or sell indicator, an indicator for

¹¹ Bonds rated AAA, AA, and A are in NAIC risk category 1 (NAIC1). NAIC2–NAIC5 correspond to BBB-, BB-, B- and CCC-rated bonds. Bonds rated CC or lower belong to NAIC6. The capital charge for NAIC1–NAIC6 is 0.4%, 1.3%, 4.6%, 10%, 23%, and 30%, respectively.

¹² Because our focus is largely on the periods surrounding Volcker Rule implementation and is less dependent on a long time series than many other corporate bond studies, we start our sample in 2006 to correspond to full TRACE implementation.

¹³ Fig. 1 considers insurance company trading, which is distinct from the trading after index exclusions by mutual funds that are considered by Dick-Nielsen and Rossi (2016). Our use of insurance company selling pressure does not preclude other episodes of selling pressure. We simply wish to identify episodes of abnormal selling pressure in which customer demand for liquidity is higher than usual.

¹⁴ Discussions with the head of fixed income investments at TIAA also reveal that trading out of bonds downgraded to junk takes about one month, on average. The exact time differs depending on the position that an insurer holds in the particular bond.

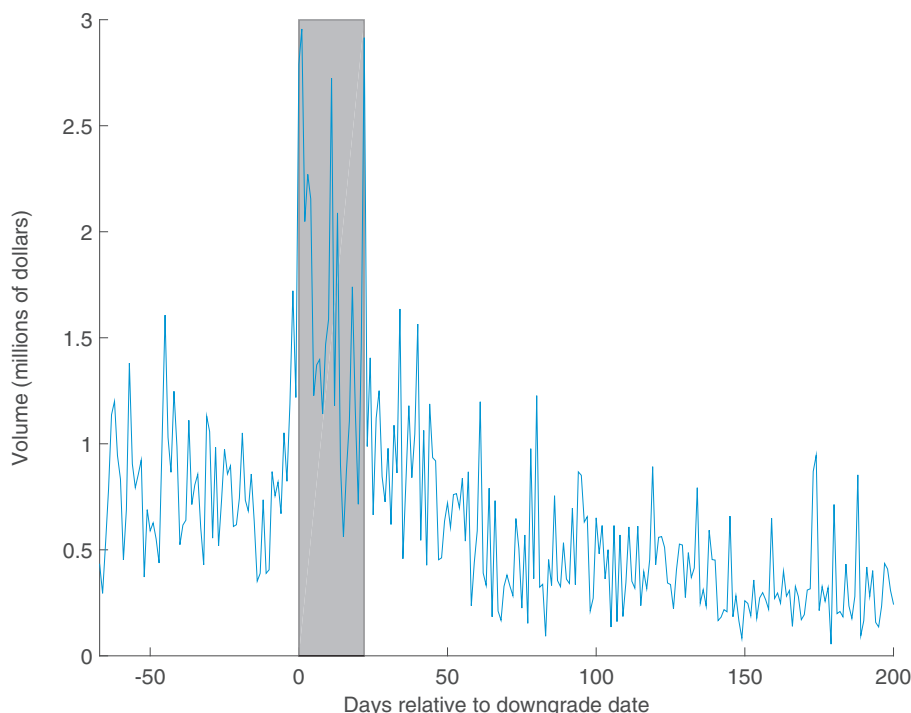


Fig. 1. Average daily trading volume for insurance companies in bonds downgraded to junk. Day 0 is the day of the first downgrade to junk. Other days are the number of weekdays relative to the downgrade date. The shaded area captures one month following the downgrade.

agency or principal trade, and an indicator for inter-dealer trade. In addition, the data contain information on dealers for each trade and for both sides of inter-dealer trades. Our version of TRACE is the regulatory version, which has dealer identities. The standard version of TRACE, while including flags for dealer–customer and inter-dealer trades, does not identify the dealer(s) involved in a trade. Knowing dealer identities allows us to separately analyze liquidity provision by Volcker-affected and non-Volcker-affected dealers. Lastly, for each of the rating downgrades in our sample, we obtain characteristics information, including total par amount outstanding, issuance date, and maturity date from Mergent FISD.

To examine bond liquidity during stress times, we estimate the average price impact during the one-month post-downgrade period in the spirit of Amihud (2002),

$$PriceImpact_i = \frac{1}{N_i - 1} \sum_{t=2}^{N_i} \frac{(P_{i,t} - P_{i,t-1})}{Q_{i,t}}, \quad (1)$$

where $P_{i,t}$ and $Q_{i,t}$ represent the price (per \$1000 of par value) and par amount (in thousands) of the t th trade in bond i , respectively, and N_i represents the total number of trades during the one month following the downgrade of bond i . Because we have transaction-level data with trade directions, we modify our calculation of the Amihud (2002) measure to use transaction-level data (as in Dick-Nielsen et al., 2012) and also use signed volume for $Q_{i,t}$, which is positive for customer buys and negative for customer sells. In calculating the price impact measure, we exclude the following transactions: when issued, canceled, subsequently corrected, reversed trades, and inter-

dealer trades.¹⁵ Following Bessembinder et al. (2009) and Ronen and Zhou (2013), we remove trades with \$100,000 or less in par amount to avoid the substantial noise that these small trades introduce into prices.

Table 1 presents our final sample of rating downgrades after matching FISD's rating history file with FINRA's TRACE data. A total of 687 bonds by 218 firms were downgraded from investment grade to speculative grade during the period from January 2006 to March 2016. Moody's acted first in 375 bonds, followed by S&P, which downgraded 247 bonds, and then Fitch, which acted first for the remaining 89 bonds. Out of the 687 bonds, 356 were downgraded by one notch and 157 were downgraded by two notches. The remaining 174 bonds were downgraded by three or more notches.

We divide our sample period into five subperiods: Pre-crisis Period (January 1, 2006–June 30, 2007), Crisis Period (July 1, 2007–April 30, 2009), Post-crisis Period (May 1, 2009–July 20, 2010), Post-Dodd–Frank Period (July 21, 2010–March 31, 2014), and Post-Volcker Period (April 1, 2014–March 31, 2016). We compare bond liquidity during the Post-Volcker Period with that during the other four subperiods prior to the effective date of the Volcker Rule, with our primary focus on comparing the Post-Volcker Period and Post-Dodd–Frank Period as we seek to ascertain

¹⁵ Inter-dealer trades are reported by both the buyer and the seller, making it impossible to identify the direction of the trade. In untabulated results, we use an Amihud measure without signed trades and include inter-dealer trades, finding that our results are similar.

Table 1

Sample description.

This table provides a description of the corporate bonds downgraded from investment grade to speculative grade by one of three major credit rating agencies [Standard & Poor's (S&P), Moody's, and Fitch] over the period from January 1, 2006 to March 31, 2016. Data on historical rating changes by the three major rating agencies are obtained from the Mergent Fixed Income Securities Database (FISD). We use the date of announcement by the rating agency that acted first to define the downgrade event. We divide the full sample period into five subperiods: Precrisis Period, Crisis Period, Post-crisis Period, Post-Dodd–Frank Period, and Post-Volcker Period. For the full sample period and each of the subperiods, we present the number of bonds downgraded and number of firms whose bonds were downgraded. We report the number of bonds in which S&P, Moody's, or Fitch was the first to take action. Multiple rating agencies can downgrade a bond on the same day. We also report the number of bonds that were downgraded by one notch, by two notches, and by three or more notches (in the "Other" column).

Period	Number of bonds	Number of firms	By S&P	By Moody's	By Fitch	By one notch	By two notches	Other
Full sample period January 1, 2006–March 31, 2016	687	218	247	375	89	356	157	174
Subperiods								
Precrisis Period (January 1, 2006–June 30, 2007)	114	45	36	50	37	54	17	43
Crisis Period (July 1, 2007–April 30, 2009)	210	57	105	100	8	97	80	33
Post-Crisis Period (May 1, 2009–July 20, 2010)	68	16	3	61	4	46	3	19
Post-Dodd–Frank Period (July 21, 2010–March 31, 2014)	113	45	51	41	33	64	28	21
Post-Volcker Period (April 1, 2014–March 31, 2016)	182	55	52	123	7	95	29	58

the additional effects of the Volcker Rule.¹⁶ The designations of the four pre-Volcker subperiods are generally consistent with other studies (e.g., Dick-Nielsen et al., 2012; Bessembinder et al., 2016).¹⁷

As pointed out by Trebbi and Xiao (2015), using exact dates of regulatory policies to study the impact of regulation on market liquidity is potentially complicated by anticipatory or delayed responses by market participants. For example, bank dealers could become more conservative in market making in anticipation of the rule prohibiting proprietary trading. In addition, regulators gave market participants over one year to fully comply with the Volcker Rule. Thus, using the effective date might capture only partial effects of the Volcker Rule on bond liquidity and biases against finding results. We address this concern with a variety of tests, but acknowledge that the complete effects (including both implementation and anticipation-related actions) could be larger than our conservative empirical methodology captures.

Table 1 shows the distribution of sample rating downgrades across the five subperiods. A total of 182 bonds were downgraded during the post-Volcker Period. The number of downgraded bonds increases from 114 for the Precrisis Period to 210 for the Crisis Period and then declines to 68 and 113 for the Post-Crisis Period and Post-Dodd–Frank Period.

4. Liquidity around stress events

Studying the effect of the Volcker Rule on corporate bond liquidity around stress events is challenging because liquidity of the bond market could have changed over time for reasons unrelated to the post-crisis regulations. To account for the potential influence of such time trends, we use a difference-in-differences methodology by first comparing the price impact in the BB bonds newly downgraded from BBB with that in the existing BB bonds and then examine how their differences have changed from the pre-Volcker periods to the Post-Volcker Period. By controlling for the baseline level of illiquidity, we also provide a control for the impact of differential macroeconomic conditions on the general level of corporate bond illiquidity. For each downgrade event, we calculate the average *PricelImpact* in bonds that were rated BB by the acting rating agency during the same one-month period, labeled as *PricelImpactControl_i*. *PricelImpactDiff_i* is the first difference and is defined as the difference in *PricelImpact* between the downgraded bond *i* and other BB bonds during the same one-month period,

$$PricelImpactDiff_i = PricelImpact_i - PricelImpactControl_i. \quad (2)$$

We then compare *PricelImpactDiff* over different periods.

¹⁶ Because our focus is on examining bond liquidity during the one month following each downgrade, we exclude those downgrade events that happened during the last month in each of the five subperiods.

¹⁷ The regulatory period that other papers study largely coincides with the period we classify as the Post-Dodd–Frank Period.

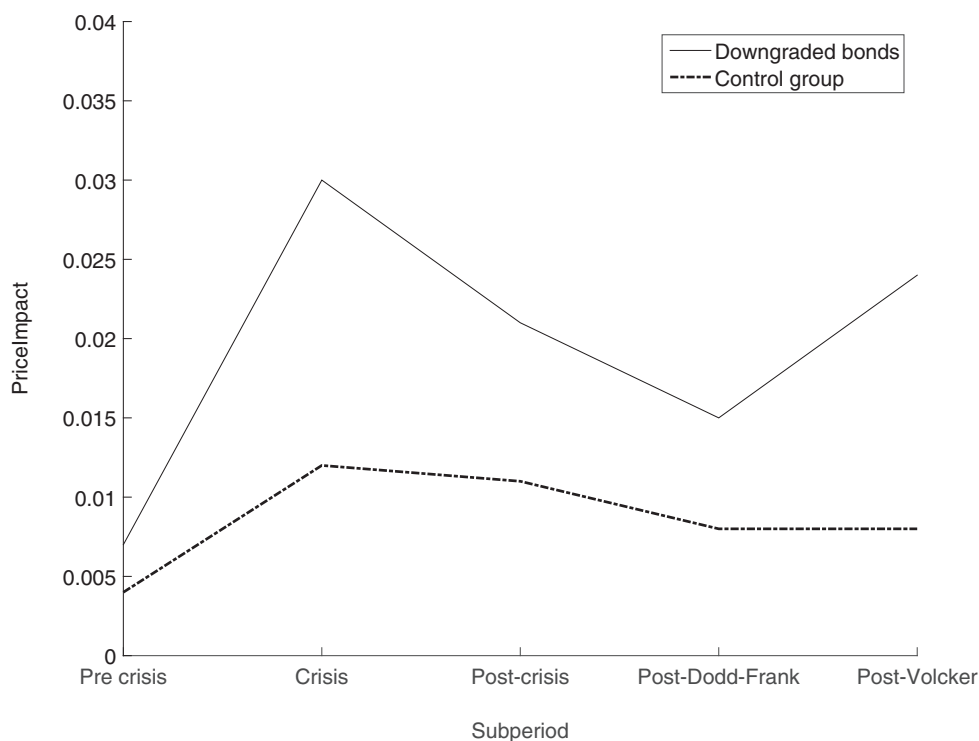


Fig. 2. Average *PriceImpact* for downgraded bonds and a control group of BB-rated bonds for the five subperiods. *PriceImpact* is as defined in Table 2.

Table 2

Univariate analysis of corporate bond liquidity following downgrades.

This table analyzes the liquidity of a bond during the one month following its downgrade from investment grade to speculative grade over the period from January 1, 2006 to March 31, 2016. Trade-level illiquidity is measured as $PriceImpactTrade = (P_t - P_{t-1})/Q_t$, where P_t and Q_t refer to the price (per \$1000 of par value) and par amount (in thousands) of the trade at time t , respectively (omitting retail-size trades, those of \$100,000 or less in par value). We then average *PriceImpactTrade* within the one month following each downgrade to get an event-level estimate. For each downgrade event, we calculate the average *PriceImpact* in bonds that were rated BB by the acting rating agency during the same one-month period (*PriceImpactControl*). *PriceImpactDiff* is *PriceImpact* – *PriceImpactControl*. The full sample is divided into five subperiods: Precrisis Period (January 1, 2006–June 30, 2007), Crisis Period (July 1, 2007–April 30, 2009), Post-crisis Period (May 1, 2009–July 20, 2010), Post-Dodd–Frank Period (July 21, 2010–March 31, 2014), and Post-Volcker Period (April 1, 2014–March 31, 2016). Because our focus is on examining bond liquidity during the one month following each downgrade, we exclude those downgrade events that happened during the last month in each subperiod. Reported *p*-values test the significance of *PriceImpactDiff*. Standard errors are clustered by firm to multiple downgrades by the same issuer.

Period	<i>PriceImpact</i>	<i>PriceImpactControl</i>	<i>PriceImpactDiff</i>	<i>p</i> -value	Number of bonds	Number of firms
Precrisis Period	0.007	0.004	0.003	0.062	114	45
Crisis Period	0.030	0.012	0.018	0.000	210	57
Post-crisis Period	0.021	0.011	0.011	0.129	68	16
Post-Dodd–Frank Period	0.015	0.008	0.007	0.000	113	45
Post-Volcker Period	0.024	0.008	0.016	0.007	182	55

4.1. Univariate analysis of price impact measures

Table 2 reports that the average *PriceImpactDiff_i* is 0.016 during the Post-Volcker Period, which is substantially higher than the mere 0.003 during the Precrisis Period. It is also higher than the 0.007 and 0.011, respectively, for the Post-Dodd–Frank Period and the Post-Crisis Period and only slightly smaller than the 0.018 for the Crisis Period. To benchmark these numbers, consider two consecutive trades at \$1000 and \$984 (per \$1000 in face value). Suppose that the second trade is a customer sell for \$1,000,000 in face value. This gives a price impact measure of 0.016 [(984 – 1000)/(–1000)].

As shown in Fig. 2, the changes in *PriceImpactDiff_i* across subperiods mainly reflect changes in *PriceImpact_i* of the downgraded bonds, instead of those of the BB bonds in the control sample. For our sample of downgraded bonds, *PriceImpact_i* was 0.007 during the Precrisis Period. It jumped to 0.03 during the Crisis Period but has since declined to 0.021 in the Post-Crisis Period and further to 0.015 in the Post-Dodd–Frank Period. However, following the implementation of the Volcker Rule, the trend of declining price impact reversed. *PriceImpact_i* increased to 0.024, higher than any of the pre-Volcker subperiods except for the Crisis Period. This finding is intriguing given that *PriceImpactControl_i* did not change from the

Post-Dodd–Frank Period to the Post-Volcker Period. In fact, the changes in $PricelImpactControl_i$ over time for the control sample of BB bonds not in stress are consistent with that shown in Bessembinder et al. (2016). Thus, focusing on bond liquidity for bonds not suffering from stress events gives an incomplete picture of the real effects of the Volcker Rule on market liquidity. In sum, bond liquidity around stress events has deteriorated since the Volcker Rule took effect.

4.2. Regression analyses

One possible explanation for the changes in bond liquidity, as shown in Table 2, is that funding conditions for dealers who provide liquidity in the market could have changed over time and this can have had an additional effect for downgraded bonds. As discussed by Brunnermeier and Pedersen (2009), poor funding conditions can lead to poor market liquidity because the ability of traders to provide market liquidity depends on their ability to obtain funding.

To address this concern, and also to control for other potentially important factors in driving changes in liquidity, we conduct a regression analysis to further assess the Volcker Rule impact. We create four dummy variables for the four subperiods after the Precrisis Period: *Crisis*, *Post-Crisis*, *Post-Dodd–Frank*, and *Post-Volcker*. *Crisis* takes the value of one if a rating downgrade occurred during the Crisis Period, and it takes the value of zero otherwise. The other three subperiod dummies are created in a similar way. We then regress $PricelImpactDiff_i$ on the four subperiod dummies and control variables. To measure funding conditions, we consider three variables. First, we follow Brunnermeier et al. (2009) and control for the Volatility Index (VIX) from the Chicago Board Options Exchange (CBOE) (VIX) and the TED spread (*TED Spread*), the difference between the three-month London Interbank Offer Rate (LIBOR) and the three-month Treasury rate, obtained from Federal Reserve Economic Data (FRED). As noted by Brunnermeier et al. (2009), higher levels of the VIX and *TED Spread* indicate tighter funding conditions. As a third measure of funding conditions, we consider the noise measure (*Noise*) from Hu et al. (2013). Their measure is based on the deviation of Treasury yields from a Svensson (1994) yield curve. The basic premise behind their measure is that when funding conditions are good, arbitrage capital is able to smooth out the yield curve and keep deviations from a smooth yield curve low.¹⁸ The Volcker Rule exempts Treasury bonds, so the ability of trading desks to arbitrage the Treasury yield curve should be unaffected by new regulations.

Although all of our sample bonds were downgraded from investment grade to speculative grade, they differ from each other in terms of both pre-downgrade rating and the number of notches downgraded. Such differences can affect bond trading following the downgrade announcement and, hence, the $PricelImpactDiff_i$ measure, so

we also include as control variables *Previous Rating* and $\Delta Rating$, which refer to the rating of the bond prior to the downgrade and the number of notches by which it was downgraded, respectively. In addition, we control for bond characteristics: (the log of) number of years since issuance (*Log Age*), number of years until maturity (*Log Time to Maturity*), and total par amount outstanding (*Log Amount Outstanding*). For ease of reference, we provide a definition of all of these independent variables in Table 3. Columns 1 and 2 of Table 4 present the result from this regression analysis. The coefficient for *Post-Volcker* is positive and significant, suggesting that illiquidity during the Post-Volcker Period is higher compared with the Precrisis Period. More importantly, the coefficient for *Post-Volcker* is significantly higher than that for *Post-Dodd–Frank*, with the magnitude being slightly larger compared with the univariate results in Table 2.

Funding conditions account for some of the variations in bond liquidity. Both VIX and *TED Spread* are significantly related to $PricelImpactDiff_i$.¹⁹ In addition, bond characteristics affect the $PricelImpactDiff_i$ measure, with older bonds and bonds with longer time to maturity experiencing lower liquidity following their downgrade, and larger issues enjoy higher liquidity. *Previous Rating* and $\Delta Rating$, which capture the potential differences in the riskiness of downgraded bonds, do not have a significant impact on the $PricelImpactDiff_i$.

To further account for the possibility that downgraded firms are fundamentally different across periods, we include as control variables a host of firm characteristics. We control for book equity divided by market equity (*B/M Ratio*) as in Bao and Hou (2017) and *Leverage*, *Total Assets*, *Interest Coverage*, and *Profit Margin* as in Han and Zhou (2014). As shown in Columns 3 and 4 of Table 4, adding these controls has little impact on the change in $PricelImpactDiff$ from the Post-Dodd–Frank Period to the Post-Volcker Period. The coefficient for *Post-Volcker* continues to be positive, and it is significantly higher than *Post-Dodd–Frank*. Our sample, however, decreases in size by 12% as this test can be conducted only on the subset of public firms with available firm characteristics.

Next, we conduct a series of additional analyses to better characterize the effect of the Volcker Rule. As an additional test on the impact of the Volcker Rule on corporate bond liquidity, we first consider pricing effects. To the extent that liquidity is priced in corporate bond yield spreads (e.g., Bao et al., 2011; Friewald et al., 2012), a deterioration in liquidity should be associated with an increase in yield spread. In Columns 1 and 2 of Table 5, we replace $PricelImpactDiff$ with $YieldSpreadDiff$, which measures the yield spread of downgraded corporate bonds relative to a control group of BB-rated corporate bonds in the month following downgrade.²⁰ Consistent with the liquidity results,

¹⁸ We obtain the noise measure through 2014 from Jun Pan's website (<http://www.mit.edu/~junpan/>) and extend the series to the end of our sample period, March 2016.

¹⁹ A higher VIX is associated with a larger $PricelImpactDiff$, though the TED spread is, surprisingly, negatively associated with $PricelImpactDiff$. The surprising sign on the TED spread is due to its very high (0.76) correlation with VIX. The insignificance of the *Noise* measure can also be attributed to its 0.92 correlation with VIX.

²⁰ We use the Treasury yield curve as the risk-free benchmark and calculate the yield spread of a corporate bond as the spread of the yield to

Table 3

Independent variable definitions. This table provides detailed definitions of the main independent variables used in the tables.

Variable	Definition
<i>Crisis</i>	Dummy variable equal to one for July 1, 2007–April 30, 2009 and zero otherwise.
<i>Post-crisis</i>	Dummy variable equal to one for May 1, 2009–July 20, 2010 and zero otherwise.
<i>Post-Dodd–Frank</i>	Dummy variable equal to one for July 21, 2010–March 31, 2014 and zero otherwise.
<i>Post-Dodd–Frank Early</i>	Dummy variable equal to one for July 21, 2010–September 30, 2013 and zero otherwise.
<i>Post-Dodd–Frank Late</i>	Dummy variable equal to one for October 1, 2013–March 31, 2014 and zero otherwise.
<i>Post-Volcker</i>	Dummy variable equal to one for April 1, 2014–March 31, 2016 and zero otherwise.
<i>Post-Volcker Early</i>	Dummy variable equal to one for April 1, 2014–July 20, 2015 and zero otherwise.
<i>Post-Volcker Late</i>	Dummy variable equal to one for July 21, 2015–March 31, 2016 and zero otherwise.
<i>Previous Rating</i>	Rating of a downgraded bond before the downgrade from investment grade to speculative grade. A numeric value is assigned to each notch of credit rating, with 1, 2, 3, and 4 ...denoting AAA, AA+, AAA, and AA- ..., respectively.
Δ Rating	Number of notches that a bond was downgraded.
<i>Log Age</i>	Log of the number of years since issuance for a bond.
<i>Log Time to Maturity</i>	Log of the number of years to maturity for a bond.
<i>Log Amount Outstanding</i>	Log of the total amount outstanding in thousands of US dollars.
<i>VIX</i>	Equity volatility index from the Chicago Board Options Exchange.
<i>TED Spread</i>	London Interbank Offered Rate (LIBOR) minus the Treasury rate from Federal Reserve Economic Data (FRED).
<i>Noise</i>	Noise measure from Hu et al. (2013).
<i>B/M Ratio</i>	Book equity divided by market equity, calculated from Compustat as $CEQ/(CSHO \times PRCC_F)$.
<i>Leverage</i>	Calculated from Compustat as $(DLC + DLTT) / (DLC + DLTT + CSHO \times PRCC_F)$.
<i>Total Assets</i>	AT from Compustat.
<i>Profit Margin</i>	OIBDP/SALE, using Compustat data.
<i>Interest Coverage</i>	$(OIADP + XINT) / XINT$, from Compustat.

Table 4

Corporate bond liquidity following downgrades.

This table analyzes how corporate bond liquidity evolves during the period from January 1, 2006 to March 31, 2016, especially following the effective date of the Volcker Rule. The dependent variable is *PriceImpactDiff*, the difference between the price impact of trading in a downgraded bond in the month after the downgrade and the average price impact for BB-rated corporate bonds in the same time horizon. Detailed definitions of all independent variables are in Table 3. Because bonds issued by the same firm are usually downgraded at the same time, we cluster the standard errors at the firm level.

Variable	Full sample		Sample with Firm characteristics	
	Estimate (1)	p-value (2)	Estimate (3)	p-value (4)
<i>Intercept</i>	−0.044	0.201	−0.016	0.392
<i>Crisis</i>	0.014	0.027	0.023	0.003
<i>Post-crisis</i>	−0.001	0.446	0.003	0.418
<i>Post-Dodd–Frank</i>	0.002	0.378	−0.002	0.335
<i>Post-Volcker</i>	0.013	0.031	0.012	0.041
<i>Previous Rating</i>	−0.001	0.386	−0.004	0.230
<i>Rating Change</i>	0.002	0.127	0.002	0.134
<i>Log Age</i>	0.005	0.011	0.006	0.012
<i>Log Time to Maturity</i>	0.009	0.009	0.008	0.007
<i>Log Amount Outstanding</i>	−0.015	0.000	−0.016	0.000
<i>VIX</i>	0.001	0.032	0.001	0.041
<i>TED Spread</i>	−0.015	0.024	−0.023	0.002
<i>Noise</i>	−0.001	0.310	−0.001	0.253
<i>B/M Ratio</i>			−0.051	0.361
<i>Leverage</i>			0.006	0.307
<i>Total Assets</i>			0.011	0.137
<i>Profit Margin</i>			0.242	0.009
<i>Interest Coverage</i>			0.050	0.053
Number of observations	687		605	
R^2	0.090		0.100	
<i>Post-Volcker versus Crisis</i>		0.424		0.110
<i>Post-Volcker versus Post-crisis</i>		0.075		0.278
<i>Post-Volcker versus Post-Dodd–Frank</i>		0.044		0.031

Table 5

Additional results.

The dependent variable in Columns 1 and 2 is *YieldSpreadDiff*, the difference between the yield spread of a downgraded bond in the month after a downgrade and the average yield spread of BB-rated bonds in the same horizon. For columns 3–8, the dependent variable is *PricelmpactDiff*. Detailed definitions of all independent variables are in Table 3. Because bonds issued by the same firm are usually downgraded at the same time, we cluster the standard errors at the firm level.

Variable	Yield spread		Post-Volcker versus Post-Dodd–Frank		Sub periods		Controlling for demand for liquidity	
	Estimate (1)	p-value (2)	Estimate (3)	p-value (4)	Estimate (5)	p-value (6)	Estimate (7)	p-value (8)
<i>Intercept</i>	−0.103	0.498	0.057	0.248	−0.043	0.208	−0.055	0.222
<i>Crisis</i>	−1.796	0.165			0.014	0.029	0.015	0.023
<i>Post-crisis</i>	−1.619	0.245			−0.001	0.468	−0.006	0.268
<i>Post-Dodd–Frank</i>	−0.151	0.465					0.001	0.405
<i>Post-Volcker</i>	2.170	0.059	0.024	0.004			0.010	0.061
<i>Post-Dodd–Frank Early</i>					0.003	0.299		
<i>Post-Dodd–Frank Late</i>					−0.008	0.095		
<i>Post-Volcker Early</i>					0.012	0.026		
<i>Post-Volcker Late</i>					0.014	0.081		
<i>Previous Rating</i>	0.068	0.484	−0.015	0.057	−0.002	0.378	−0.002	0.404
<i>Rating Change</i>	0.853	0.043	0.001	0.380	0.002	0.138	0.001	0.291
<i>Log Age</i>	1.004	0.016	0.002	0.292	0.006	0.011	0.005	0.018
<i>Log Time to Maturity</i>	−3.069	0.000	0.016	0.001	0.009	0.009	0.010	0.007
<i>Log Amount Outstanding</i>	1.055	0.014	−0.024	0.004	−0.015	0.000	−0.016	0.002
<i>VIX</i>	0.236	0.121	0.005	0.004	0.001	0.036	0.002	0.022
<i>TED Spread</i>	3.490	0.140	−0.131	0.007	−0.014	0.026	−0.019	0.001
<i>Noise</i>	0.068	0.447	−0.009	0.084	−0.001	0.327	−0.001	0.248
<i>Issuer Stock CAR</i>							0.005	0.324
<i>Log Insurer Trade</i>							−0.001	0.136
<i>Log All Downgrade Bonds</i>							0.001	0.257
Number of observations	687		295		687		613	
R ²	0.501		0.129		0.091		0.098	
Post-Volcker versus Crisis		0.074						0.258
Post-Volcker vs Post-crisis		0.004						0.071
Post-Volcker vs Post-Dodd Frank		0.004						0.099
Post-Volcker Early vs Crisis						0.383		
Post-Volcker Early vs Post-Crisis						0.103		
Post-Volcker Early versus Post-Dodd–Frank Early						0.081		
Post-Dodd–Frank Late versus Post-Dodd–Frank Early						0.025		
Post-Volcker Early versus Post-Dodd–Frank Late						0.004		
Post-Volcker Late versus Post-Volcker Early						0.420		

we find that *YieldSpreadDiff* has increased from the Post-Dodd–Frank Period to the Post-Volcker Period, suggesting a greater price impact of downgrades after Volcker Rule implementation.

As an alternative way to isolate Volcker Rule effects, we focus solely on the sample of downgraded bonds in the Post-Dodd–Frank Period and the Post-Volcker Period, during which funding conditions were relatively stable. In Columns 3 and 4 of Table 5, the *Post-Volcker* dummy measures the changes in liquidity from the Post-Dodd–Frank Period. We find an even stronger effect of the Volcker Rule in this specification.

In addition, we examine how liquidity has evolved within each subperiod around the Volcker Rule implementation. We split up the Post-Dodd–Frank Period into Post-Dodd–Frank Early and Post-Dodd–Frank Late with Post-Dodd–Frank Late covering the last six months of the Post-Dodd–Frank Period (i.e., from October 1, 2013–March 31, 2014). For the Post-Volcker Period, we choose as a cut-off July 21, 2015, the date when full conformance with

the Volcker Rule was required. Comparing the two post-Volcker periods with the last six months of the Post-Dodd–Frank Period, we find even stronger results than for our base regression. Liquidity improves from the early part of the Post-Dodd–Frank Period to the late part of the Post-Dodd–Frank Period. If dealers initiated compliance with the Volcker Rule prior to the conformance period, liquidity would likely begin to decline in the few months leading up to Volcker Rule implementation. We find the opposite. The difference between the two post-Volcker periods is insignificant. Duffie (2012) argues that, in the long run, non-Volcker dealers and nonbanks are likely to step in and provide the liquidity that Volcker-affected dealers previously provided.²¹ Our results show that such change has not yet occurred during the period we study in this paper.

maturity on the corporate bond over the yield to maturity on a default-free bond with the same time to maturity.

²¹ Duffie (2012) argues that such a migration outside the regulated bank sector could have unintended negative consequences for financial stability. Here, we do not address the overall costs and benefits of such a change and solely examine whether non-Volcker dealers have already picked up enough market making to allow illiquidity around downgrades to return to previous levels.

Another potential explanation for our finding is that demand for liquidity changed after the Volcker Rule. For example, some variation in information could exist during the month following a downgrade that is associated with different demand for liquidity. If this is the case, then information about the downgraded firm is driving the reluctance of dealers in providing liquidity instead of a fear of violating the Volcker Rule. In addition, one could expect that the amount of insurer trading or the amount of bonds downgraded at the same time has an impact on liquidity provision. To address this concern, we include three additional variables as controls. We control for the issuer's cumulative abnormal return (CAR) as the equity return of the issuer from one day before the downgrade to one month after the downgrade minus the Center for Research in Security Prices value-weighted index during the same window.²² We also control for the total amount of trading by insurance companies during the one month post-downgrade window, as well as the total par amount of other bonds downgraded during the one month around the downgrade. Columns 7 and 8 of Table 5 show that none of these variables is significantly related to *PricelmpactDiff* after we control for funding conditions and bond characteristics. We continue to observe a similar increase in *PricelmpactDiff* after the Volcker Rule. The increase has weaker statistical significance, partly because of the drop in sample size due to missing information for these additional controls.

4.3. Robustness checks

We conduct a number of tests to check the robustness of our results. First, we consider two alternative liquidity measures: realized bid-ask spreads and a lower frequency (Amihud, 2002) measure. Realized bid-ask spreads are measured as the average daily difference between volume-weighted average customer buy prices (*Ask*) and volume-weighted average customer sell price (*Bid*) during the one month following the downgrade. The lower frequency Amihud measure is defined as

$$Amihud_{IT} = \frac{1}{N_t} \sum_{t \in T} \frac{|P_t - P_{t-1}|}{Volume_t}, \quad (3)$$

where N_T is the number of days traded in month T , P_t denotes the closing price of a bond on day t , and $Volume_t$ is the total trading volume on day t . As shown in Columns 1–4 of Table 6, our conclusions are generally unchanged.

Second, we two-way cluster standard errors (Petersen, 2008; Cameron et al., 2011) by month of downgrade and firm, which allows for standard errors to be clustered across all downgrades in a month, including downgrades by different issuers, and also across time for a firm. If anything, we find slightly stronger statistical significance (see Columns 5 and 6), likely due to negative correlation of errors within downgrade months.

Finally, we use a matched sample approach by comparing each downgraded BB bond with a subsample of the BB bonds that are similar to the downgraded BB bond in terms of time to maturity, total par amount outstanding, and age. We first segment BB bonds in the control group into three time to maturity categories: short term (maturing within one year), medium term (with time to maturity greater than one year but no more than seven years), and long term (maturing in over seven years). Within each maturity category, we further segment bonds into three size categories: small issue, medium issue, and large issue, using \$0.5 billion and \$1.5 billion in total par amount outstanding as the cutoffs. Finally, we divide bonds within each size category into new issues and seasoned issues, depending on whether its time since issuance is greater than one year. Therefore, we form a total of 18 bond groups in the control sample based on time to maturity, amount outstanding, and age. We then calculate *PricelmpactDiff* for each downgraded bond by taking the difference between the *Pricelmpact* of the downgraded bond and the average *Pricelmpact* of BB bonds from the matching group during the same one-month period. Columns 7 and 8 of Table 6 show that using the matched sample approach has little impact on our results. We continue to observe that, following implementation of the Volcker Rule, the marginal deterioration in bond liquidity during stress times has significantly increased compared with the period just before Volcker Rule implementation.

5. Dealer behavior around stress events

In this section, we turn to the question of why liquidity has changed following the implementation of the Volcker Rule. We study how dealer behavior changed around Volcker Rule implementation and, importantly, compare the behavior of Volcker-affected dealers and non-Volcker-affected dealers. In Section 5.1, we discuss how we identify whether a dealer is Volcker-affected, and in Sections 5.2 and 5.3, we show the change in behavior across the two groups over time. Finally, we discuss other regulations, in Section 5.4.

5.1. Identifying Volcker-affected dealers

A key issue is identifying which broker-dealers are subject to the Volcker Rule. This is a nontrivial task as full lists of Volcker-affected institutions are not published. In a study of the Volcker Rule, (Oliver Wyman and SIFMA, 2011) categorize a list of 21 liquidity providers into those affected or unaffected by the Volcker Rule. (The list is reproduced in the Appendix.) They identify four (Cantor Fitzgerald & Co., Daiwa Capital Markets, Jefferies & Co., and Nomura) that are not affected by the Volcker Rule. Among those affected are major bank holding companies (BHCs) such as Goldman Sachs and Morgan Stanley. However, this list is far from complete as TRACE data identify hundreds of dealers transacting in the bond market.

To determine whether other broker-dealers are covered by the Volcker Rule, we follow the principle that the Volcker Rule was designed to prevent institutions with access to government backstops from participating in proprietary

²² See Ambrose et al. (2012) for a discussion of the possible information effects surrounding a downgrade. We also experiment with shorter windows such as a $[-1, +1]$ window as in Bao and Edmans (2011) and windows from one day before downgrade to one week and two weeks after the downgrade. Results are similar.

Table 6

Robustness checks.

The dependent variable is *PricelmpactDiff*. In Columns 1 and 2, it is calculated using the bid-ask spread. In Columns 3 and 4, it is calculated using a lower frequency Amihud measure, based on daily observations instead of transaction-level data. In Columns 5 and 6, the calculation of *PricelmpactDiff* is the same as in Table 4, but standard errors are two-way clustered by firm and month of downgrade. In Columns 7 and 8, *PricelmpactDiff* is calculated as in Table 4, but with the control group being matched by time to maturity, age, and amount outstanding, in addition to having a BB rating. Detailed definitions of all independent variables are in Table 3. Because bonds issued by the same firm are usually downgraded at the same time, we cluster the standard errors at the firm level.

Variable	Spread		Low frequency Amihud		Cluster by firm and month		Matched sample	
	Estimate (1)	p-value (2)	Estimate (3)	p-value (4)	Estimate (5)	p-value (6)	Estimate (7)	p-value (8)
<i>Intercept</i>	−0.398	0.053	0.000	0.500	−0.044	0.447	−0.045	0.190
<i>Crisis</i>	0.075	0.133	0.021	0.014	0.014	0.039	0.016	0.020
<i>Post-crisis</i>	0.022	−0.388	0.004	0.338	0.001	0.885	0.000	0.488
<i>Post-Dodd–Frank</i>	0.033	0.296	0.001	0.446	0.002	0.713	0.002	0.388
<i>Post-Volcker</i>	0.075	0.099	0.017	0.014	0.013	0.008	0.013	0.032
<i>Previous Rating</i>	0.023	−0.151	0.008	0.054	−0.001	0.771	−0.001	0.425
<i>Rating Change</i>	−0.021	0.002	0.000	0.415	0.002	0.229	0.001	0.260
<i>Log Age</i>	−0.008	0.287	0.009	0.000	0.005	0.005	0.005	0.020
<i>Log Time to Maturity</i>	0.073	0.002	0.014	0.000	0.009	0.015	0.009	0.006
<i>Log Amount Outstanding</i>	−0.001	0.473	−0.022	0.000	−0.015	0.001	−0.015	0.000
<i>VIX</i>	0.002	0.324	0.001	0.054	0.001	0.043	0.001	0.065
<i>TED Spread</i>	0.009	−0.433	0.027	0.002	−0.015	0.009	−0.014	0.034
<i>Noise</i>	0.005	−0.280	0.000	0.480	0.001	0.593	−0.001	0.308
Number of observations	647		687		687		687	
R ²	0.056		0.155		0.090		0.077	
<i>Post-Volcker versus Crisis</i>		0.497		0.361		0.818		0.349
<i>Post-Volcker versus Post-crisis</i>		0.119		0.097		0.133		0.095
<i>Post-Volcker versus Post-Dodd–Frank</i>		0.049		0.012		0.035		0.044

trading. The two most prominent backstops mentioned in the *Federal Register* discussion of the Volcker Rule are FDIC insurance and access to the Fed's discount window (*Federal Register*, 2014). We start with the broker-dealers on the (Oliver Wyman and SIFMA, 2011) list and add to it the most active three hundred broker-dealers in terms of trading volume. Together, these broker-dealers account for 97% of total bond market trade volume. We then search both the FDIC's database of FDIC-insured banking institutions and the National Information Center (NIC) institution database to see which of the three hundred broker-dealers were subject to the Volcker Rule. If a broker-dealer, or more likely an affiliate (i.e., a commercial bank with the same parent holding company) is listed as having FDIC insurance, we code it as Volcker-affected. The NIC data is more complicated as it contains "banks and other institutions for which the Federal Reserve has a supervisory, regulatory, or research interest." Thus, not all institutions in the database are necessarily Volcker-affected. We look for institutions coded as national banks, state member banks, bank holding companies, and financial holding companies and treat these as Volcker-affected. Among the main types of institutions in the NIC database that we do not treat as Volcker-affected are securities broker-dealers and domestic entity other. We also search the Federal Reserve Board's Resolution Plans website (<https://www.federalreserve.gov/bankinforeg/resolution-plans.htm>) to identify large bank holding companies under Fed supervision that must submit a living will.²³ Combining results from our search with

the list in Oliver Wyman and SIFMA (2011) results in approximately 45% of the most active dealers being determined to be Volcker-affected.

5.2. Dealer trading activities

We start by showing basic dealer trading patterns around downgrade-to-speculative (stress) events in Table 7. In the month following a downgrade, the average turnover of downgraded bonds is close to 40% in the Post-Volcker Period, higher than any of the other four periods in our sample. It is also much higher than the 9% monthly turnover of BB bonds during the Post-Volcker Period, consistent with significant selling by insurance companies in the period just after downgrades to speculative grade. However, the proportion of total trading volume that is dealer-customer (as opposed to inter-dealer), 62%, is roughly in line with the other periods of our sample.

Of perhaps more interest, we compare Volcker-affected and non-Volcker-affected dealers in Panels B and C of Table 7. Our focus is on the proportion of dealer-customer trading handled by Volcker versus non-Volcker dealers and also the dealers' use of agency trading. The underlying evidence in Ellul et al. (2011) is that, around stress events, some institutional investors (e.g., insurance companies) sell bonds due to regulatory constraints. Dealers then intermediate these trades and potentially hold inventory in bonds when selling demand exceeds buying demand. Both the proportion of customer-dealer trades and the percentage of agency trades address how dealers react to customer demands.

Volcker-affected dealers tend to be larger than non-Volcker dealers and handled 93% of dealer-customer

²³ Living wills are mandated by Dodd–Frank to prevent taxpayer bailouts in the future. Thus, the fact that regulators require a living will suggests that these are institutions with government backstops.

Table 7

Trading activity by dealers affected by the Volcker Rule and those not affected by the Volcker Rule.

This table presents summary information on trading activities by dealers affected by the Volcker Rule and those not affected by the Volcker Rule during each of the five subperiods between January 1, 2006 and March 31, 2016: Precrisis Period (January 1, 2006–June 30, 2007), Crisis Period (July 1, 2007–April 30, 2009), Post-crisis Period (May 1, 2009–July 20, 2010), Post-Dodd–Frank Period (July 21, 2010–March 31, 2014), and Post-Volcker Period (April 1, 2014–March 31, 2016). *Turnover* is the aggregate trading volume by all dealers, including both inter-dealer trades and dealer–customer trades, during the month following each rating downgrade, scaled by amount outstanding. *Percent of volume that is dealer–customer* measures the proportion of trading volume that is dealer–customer trading (as opposed to inter-dealer trading). *Share of dealer–customer trade* reports the percentage of total dealer–customer trading that is either attributable to Volcker-affected dealers (Panel B) or non-Volcker dealers (Panel C). *Percent of trades that are agency* measures the proportion of trades that are effectively agency trades. We classify a trade as being effectively agent if it is offset by another trade that occurred within one minute with the same trade size by the same dealer but with opposite trade direction. This one-minute algorithm is similar to that used in [Harris \(2015\)](#) and [Bessembinder et al. \(2016\)](#).

Panel A: Full sample trading statistics			
Period	Turnover	Percent of volume that is dealer–customer	
Precrisis Period	0.300	65.601	
Crisis Period	0.277	70.037	
Post-crisis Period	0.219	62.324	
Post-Dodd–Frank Period	0.306	53.913	
Post-Volcker Period	0.383	62.032	
Panel B: Volcker-affected dealers			
Period	Percent of volume that is dealer–customer	Share of dealer–customer trade	Percent of trades that is agency
Precrisis Period	77.428	93.371	12.104
Crisis Period	83.295	89.117	15.413
Post-crisis Period	75.958	84.569	15.543
Post-Dodd–Frank Period	67.224	79.728	15.965
Post-Volcker Period	75.608	76.297	22.709
Panel C: Non-Volcker dealers			
Period	Percent of volume that is dealer–customer	Share of dealer–customer trade	Percent of trades that is agency
Precrisis Period	22.119	6.629	46.404
Crisis Period	37.803	10.883	45.326
Post-crisis Period	29.192	15.431	35.779
Post-Dodd–Frank Period	33.063	20.272	33.118
Post-Volcker Period	48.722	23.703	29.403

volume around stress events in the Precrisis Period. Thus, of the three hundred dealers in our sample, the most active tend to be those that are subject to the Volcker Rule. Given the relatively large market share of Volcker-affected dealers, this already suggests that non-Volcker dealers were unlikely to have stepped in and provided enough additional liquidity to offset any changes by Volcker-affected dealers. Over our sample period, a gradual decline is evident in the share of dealer–customer volume handled by Volcker-affected dealers. By the Post-Volcker Period, non-Volcker dealers were handling almost one-quarter of the dealer–customer volume. Though the increasing volume handled by non-Volcker dealers is consistent with Volcker-affected dealers scaling back their market making due to the Volcker Rule, we cannot rule out the explanation that a gradual time series change occurred in the dealer business that has led the smaller, non-Volcker dealers to take a greater share of dealer–customer volume.

Next, we turn to how agency trading has changed over time for Volcker and non-Volcker dealers. Agency trading occurs when a dealer has lined up a counterparty to immediately offset a trade with a customer. For example, if an insurance company decides to sell a downgraded bond, a dealer in an agency trade would line up another customer (or dealer) to purchase the bond. In such a case, the dealer does not commit capital or take on any inventory risk. A principal trade, in contrast, involves a dealer that

takes on one side of a trade without preexisting knowledge that it will be able to unwind the trade quickly. We follow [Harris \(2015\)](#) and [Bessembinder et al. \(2016\)](#) and define a trade as an agency trade if it is offset by another trade in the opposite direction within one minute. Agency trades are particularly attractive to dealers concerned about the Volcker Rule as they appear to regulators as obvious market making; that is, they have no inventory aging, they exhibit extremely fast turnover, and all profits are from bid-ask spreads and commissions.

Our empirical results indicate that Volcker-affected dealers have increased the proportion of their total volume done on an agency basis. Precrisis, only 12% of the volume traded by Volcker-affected dealers was in agency trades. This number jumped to a little over 15% with the onset of the 2007–2009 financial crisis and stayed fairly flat until jumping again to almost 23% with enforcement of the Volcker Rule. The sudden jump in the proportion of volume done as agency trades is suggestive of a causal effect of the Volcker Rule on Volcker-affected dealers' willingness to hold bonds on their balance sheet without prearranging an offsetting trade. Non-Volcker dealers, in contrast, have seen a decline in the proportion of trades that they do on an agency basis. During the Precrisis Period, almost half of the trades done by non-Volcker dealers around stress events were done as agency trades. By the Post-Volcker Period, this percentage had dropped to 29%.

Table 8

Volcker Rule and agency trades.

This table analyzes how dealers' willingness to arrange trades on a principal basis changes following the Volcker Rule. For Columns 1, 2 and 3, 4, the dependent variable is the proportion of dealer–customer traded completed on effectively agent basis (expressed in decimals) for Volcker-affected dealers and non-Volcker-affected dealers, respectively. The primary independent variables of interest are dummy variables for *Crisis*, *Post-crisis*, *Post-Dodd–Frank*, and *Post-Volcker*. Detailed definitions of all independent variables are in Table 3. Because bonds issued by the same firm are usually downgraded at the same time, we cluster the standard errors at the firm level.

Variable	Dealers affected by Volcker Rule		Dealers not affected by Volcker Rule	
	Estimate (1)	p-value (2)	Estimate (3)	p-value (4)
<i>Intercept</i>	0.250	–0.147	0.180	0.292
<i>Crisis Period</i>	0.001	–0.494	0.114	0.070
<i>Post-crisis Period</i>	0.085	–0.065	0.098	0.090
<i>Post-Dodd–Frank Period</i>	0.085	–0.004	0.043	0.271
<i>Post-Volcker Period</i>	0.144	0.000	–0.087	0.089
<i>Previous Rating</i>	–0.014	0.267	0.045	0.066
<i>Rating Change</i>	–0.007	0.169	0.015	0.096
<i>Log Age</i>	0.014	0.127	0.022	0.123
<i>Log Time to Maturity</i>	–0.001	0.483	–0.004	0.440
<i>Log Amount Outstanding</i>	–0.036	0.004	–0.023	0.174
<i>VIX</i>	–0.008	0.003	0.000	0.462
<i>TED Spread</i>	0.040	0.064	0.010	0.419
<i>Noise</i>	0.025	0.001	0.015	0.168
Number of observations	687		687	
R^2	0.067		0.036	
<i>Post-Volcker versus Crisis</i>		0.000		0.349
<i>Post-Volcker versus Post-crisis</i>		0.137		0.406
<i>Post-Volcker versus Post-Dodd–Frank</i>		0.036		0.172

To more formally study the changes in agency trades across time for Volcker and non-Volcker-affected dealers, we run a regression of proportion of trades that are agency trades on period dummies and controls. Our base regression is

$$\begin{aligned} & \text{Proportion of agency volume}_i \\ &= \beta_0 + \beta_1 \text{Crisis} + \beta_2 \text{Post-Crisis} \\ &+ \beta_3 \text{Post-Dodd-Frank} + \beta_4 \text{Post-Volcker} + \gamma X + \epsilon, \quad (4) \end{aligned}$$

where the unit of observation is a stress event, the dependent variable is the proportion of volume done by either Volcker or non-Volcker dealers on an agency basis, and the omitted period dummy is *Pre-crisis*. X represents a vector of control variables that are the same as defined in Table 3 and used in Table 4. Our variable of interest is β_4 , which directly measures the difference in the proportion of volume done on an agency basis between the Post-Volcker Period and the Pre-crisis Period. Also of particular interest is the difference between β_4 and the coefficients on the other subperiod dummies, particularly $\beta_4 - \beta_3$.

The regression results are presented in Table 8. In Columns 1 and 2, the dependent variable is the proportion of agency trading done by Volcker Rule – affected dealers after stress events. The coefficient on the *Post-Volcker* dummy is 0.144, indicating a 14 percentage point increase in volume done on an agency basis relative to the pre-crisis period. This change is slightly larger than the 11 percentage point increase without controls in Table 7. Importantly, the coefficient on the *Post-Volcker* dummy is also significantly

larger than for the other periods in our sample. The Post-Dodd–Frank Period has a statistically significant coefficient of 0.085, indicating an 8.5 percentage point increase in agency trades compared with the Pre-crisis Period, but it is also much smaller than the Post-Volcker Period. The 6 percentage point increase in agency trading from the Post-Dodd–Frank Period to the Post-Volcker Period is both statistically and economically significant.

In Columns 3 and 4 of Table 8, we rerun our agency trade regression but instead consider the proportion of trades done on an agency basis by non-Volcker-affected dealers. While the coefficients on all of the subperiod dummies are negative and the amount of agency trading done by non-Volcker dealers is smaller (in magnitude) during the Post-Volcker Period as compared with the Post-Dodd–Frank Period, we do not find any statistical significance. Unlike Volcker-affected dealers, a sharp jump in the proportion of agency trading for non-Volcker dealers upon implementation of the Volcker Rule is not evident. If anything, we find the opposite, at least in terms of point estimates. Our results are consistent with the Volcker Rule inducing Volcker-affected dealers to shift from principal to agency trading as a way to avoid inventory imbalance.

5.3. Dealer capital commitment

A more direct measure of dealers' willingness to hold inventory imbalances is the time-weighted capital commit-

ment.²⁴ In the one month following a stress event, we calculate for each dealer the absolute deviation from starting inventory. The intuition is that if a dealer starts with a particular desired inventory level, the first purchase moves the dealer above this desired inventory level, but a following sell moves the dealer back toward the desired inventory level. The desired inventory level is unobservable, so our implicit assumption is that the starting level of inventory is optimal. One could worry that dealers pre-position prior to downgrades and that the level of pre-positioning has changed since Volcker Rule implementation. However, we find little evidence of systematic pre-positioning by dealers. We first calculate dealer net trading by subtracting dealer sell volume from dealer buy volume (both in par amount) during the month prior to each downgrade. We then divide it by the bond's outstanding par amount. Dealer net trading prior to rating downgrade has not significantly changed (both economically and statistically) following the Volcker Rule implementation.²⁵

To calculate how far a dealer is from the starting inventory level, we simply take the accumulated dollar amount bought and subtract the accumulated dollar amount sold from the starting point. To calculate the time-weighted capital commitment, we average the absolute distance from the starting inventory, weighting by the amount of time the inventory level is held. Essentially, the measure captures the amount of order imbalance that a dealer is willing to accept.

While our measure is similar to the dealer capital commitment measure in Bessembinder et al. (2016), we consider capital commitment over the course of a month, while they construct a daily measure. Their measure implicitly assumes that the starting point for each day is the optimal inventory, whereas our monthly measure allows for inventory to continue to move away from optimal inventory over the course of a few days.²⁶ If a dealer has purchased a large volume of a bond in a day and has not sold this volume to another customer or dealer, the dealer still has significant capital commitment the next day. Once we calculate bond-dealer – level capital commitment, we sum across dealers for a given stress event.

In Table 9, we report regressions of time-weighted dealer capital commitment on subperiod dummies and controls separately for Volcker-affected and non-Volcker-affected dealers, similar to our proportion of agency volume regressions. In Columns 1 and 2, we show that dealer capital commitment by Volcker-affected dealers has declined in all periods relative to the Precrisis Period. Dealer capital commitment is roughly \$5 million in face value lower on average for a downgraded bond during the Crisis Period and \$8 million lower for the Post-Crisis Period

and Post-Dodd–Frank Period as compared with the Precrisis Period.²⁷ For the Post-Volcker Period, this decline is \$16 million in face value relative to the Precrisis Period. The decline in capital commitment for the Post-Volcker Period is also statistically larger than the Crisis Period, Post-Crisis Period, and Post-Dodd–Frank Period. Thus, while a large and sudden drop in capital commitment is evident from the Precrisis Period to the Crisis Period, a large and sudden drop also exists from the Post-Dodd–Frank Period to the Post-Volcker Period, suggesting that a significant shift occurred in Volcker-affected dealers around the implementation of the Volcker Rule. Columns 3 and 4 of Table 9 consider whether capital commitment has changed for non-Volcker dealers. Our results indicate a different pattern for non-Volcker dealers as they have a jump in capital commitment in the Post-Dodd–Frank Period that they maintain through the Post-Volcker Period. However, the coefficient of *Post-Volcker* for Volcker-affected dealers is about ten times as large as that for non-Volcker-affected dealers, suggesting that any additional liquidity provided by the non-Volcker-affected dealers is not sufficient to offset the drop by Volcker-affected dealers.

5.4. Capital commitment and Basel III

A potential concern in trying to isolate a Volcker Rule implementation effect is that, in the post-crisis period, a number of reforms were passed to regulate the finance industry. The Comprehensive Capital Analysis and Review process began in 2011, requiring bank holding companies to submit capital plans to the Federal Reserve. The capital plan requires that the BHC be able to maintain minimum capital requirements even under stress scenarios, providing a stiff test of a BHC's regulatory capital. The punishment for not passing a CCAR test is that the BHC is not allowed to make capital distributions unless the Federal Reserve indicates in writing that it allows the distribution.²⁸ Each year, the Federal Reserve publishes a list of BHCs that have either failed their CCAR tests or received only a conditional pass.

In January 2014, the start of Basel III implementation added capital requirements above what was required in Basel II. In conjunction with CCAR regulations, this potentially made banks more capital constrained and could have caused BHCs to change their market-making businesses.²⁹ To test the hypothesis that the combination of Basel III and CCAR is driving our results on dealers, we split dealers into those that were CCAR-constrained and those that were not. We classify any bank that failed a CCAR test or was given a conditional pass in 2014 or 2015 as CCAR-constrained.³⁰

²⁴ While agency trades are desirable trades for dealers worried about the Volcker Rule, the proportion of agency trades does not fully capture changes in dealer capital commitment. They do not capture the effect of dealers shortening the time that they commit capital for trades that are not agency trades.

²⁵ We also consider a similar exercise on dealer net trading of similarly rated bonds, finding that net trading prior to rating downgrades for these bonds has not significantly changed after Volcker Rule implementation.

²⁶ Their measure's assumption seems particularly unlikely to be true around bond downgrades when order imbalance tends to be persistent.

²⁷ As a benchmark, the average capital commitment of Volcker-affected dealers in BB-rated bonds during non-stress periods is \$6 million. For non-Volcker dealers, it is \$1.6 million.

²⁸ Historically, the Federal Reserve has continued to allow failed BHCs to continue capital distributions at the same rate as in the past. Effectively, the main constraint is that these BHCs cannot increase their capital distributions.

²⁹ We thank Darrell Duffie for suggesting the CCAR linkage to us.

³⁰ BHCs receiving a conditional pass are required to remediate deficiencies and resubmit a new capital plan later in the year. Thus, such BHCs would have similar incentives to BHCs that fail CCAR tests.

Table 9

Dealer capital commitment around Volcker Rule implementation.

This table analyzes how dealers' willingness to commit their own capital to bond trading changes following the Volcker Rule. For Columns 1, 2 and 3, 4, the dependent variable is the time-weighted capital commitment following a rating downgrade for Volcker-affected and non-Volcker-affected dealers, respectively. The primary independent variables of interest are dummy variables for *Crisis*, *Post-crisis*, *Post-Dodd–Frank*, and *Post-Volcker*. Detailed definitions of all independent variables are in Table 3. Because bonds issued by the same firm are usually downgraded at the same time, we cluster the standard errors at the firm level.

Variable	Dealers affected by Volcker Rule		Dealers not affected by Volcker Rule	
	Estimate (1)	p-value (2)	Estimate (3)	p-value (4)
<i>Intercept</i>	34032.234	0.018	6728.739	0.002
<i>Crisis Period</i>	−5159.552	0.060	−265.744	0.139
<i>Post-crisis Period</i>	−8594.994	0.004	578.988	0.069
<i>Post-Dodd–Frank Period</i>	−8289.824	0.005	2017.328	0.000
<i>Post-Volcker Period</i>	−16256.224	0.000	1958.222	0.000
<i>Previous Rating</i>	331.105	0.412	−434.793	0.016
<i>Rating Change</i>	53.393	0.451	−80.834	0.133
<i>Log Age</i>	−3098.077	0.001	−298.886	0.046
<i>Log Time to Maturity</i>	2334.237	0.018	33.220	0.418
<i>Log Amount Outstanding</i>	12546.038	0.000	995.744	0.000
<i>VIX</i>	−266.544	0.034	−9.777	0.367
<i>TED Spread</i>	373.787	0.412	127.922	0.280
<i>Noise</i>	−217.906	0.258	−13.179	0.417
Number of observations	687		687	
R ²	0.384		0.209	
<i>Post-Volcker versus Crisis</i>		0.000		0.000
<i>Post-Volcker versus Post-crisis</i>		0.000		0.152
<i>Post-Volcker versus Post-Dodd–Frank</i>		0.000		0.531

Table 10

Capital Commitment by Volcker-Affected Dealers: The Effect of Comprehensive Capital Analysis and Review Testing.

This table analyzes how CCAR regulations affect capital commitment among Volcker-affected dealers. For Columns 1, 2 and 3, 4, the dependent variable is the time-weighted capital commitment following a rating downgrade for Volcker-affected dealers who passed the CCAR test and those failed or conditionally passed the CCAR test, respectively. The primary independent variables of interest are dummy variables for *Crisis*, *Post-crisis*, *Post-Dodd–Frank*, and *Post-Volcker*. Detailed definitions of all independent variables are in Table 3. Because bonds issued by the same firm are usually downgraded at the same time, we cluster the standard errors at the firm level.

Variable	Volcker-affected dealers who passed the CCAR test		Volcker-affected dealers who failed/conditionally passed the CCAR test	
	Estimate (1)	p-value (2)	Estimate (3)	p-value (4)
<i>Intercept</i>	16068.380	0.044	17963.854	0.019
<i>Crisis Period</i>	−2715.693	0.050	−2443.859	0.160
<i>Post-crisis Period</i>	−3848.968	0.016	−4746.027	0.004
<i>Post-Dodd–Frank Period</i>	−5817.876	0.000	−2471.948	0.103
<i>Post-Volcker Period</i>	−8527.929	0.000	−7728.295	0.000
<i>Previous Rating</i>	−454.329	0.305	785.433	0.155
<i>Rating Change</i>	−14.287	0.476	67.680	0.397
<i>Log Age</i>	−1624.691	0.001	−1473.387	0.010
<i>Log Time to Maturity</i>	1525.766	0.001	808.471	0.137
<i>Log Amount Outstanding</i>	3296.999	0.000	9,249.040	0.000
<i>VIX</i>	22.233	0.372	−288.777	0.005
<i>TED Spread</i>	−492.810	0.262	866.597	0.227
<i>Noise</i>	−420.737	0.006	202.831	0.220
Number of observations	687		687	
R ²	0.273		0.341	
<i>Post-Volcker versus Crisis</i>		0.000		0.019
<i>Post-Volcker versus Post-crisis</i>		0.000		0.015
<i>Post-Volcker versus Post-Dodd–Frank</i>		0.003		0.000

If BHCs change their market-making behavior in response to changing capital requirements, then CCAR-constrained banks would be expected to lower their capital commitment more than BHCs that were able to pass their CCAR tests.

Dealers that passed the CCAR tests and dealers that failed or conditionally passed the CCAR test had lower capital commitment following rating downgrades in the Volcker Rule implementation period as compared with the Post-Dodd–Frank Period (see Table 10). For both groups, a significant decline in capital commitment is evident during the Post-Volcker Period relative to both the Precrisis Period and the Post-Dodd–Frank Period. Hoarding capital to pass the CCAR test is thus not supported as an explanation for decreased dealer capital commitment in bond trading.

6. Conclusion

In this paper, we study the impact of Volcker Rule implementation on corporate bond illiquidity and dealer behavior. Our main finding is that the Volcker Rule has a deleterious effect on corporate bond liquidity and that dealers subject to the rule become less willing to provide liquidity during stress times. While weak evidence exists that dealers not affected by the Volcker Rule have stepped in to provide some liquidity, we find that the net effect is a less liquid corporate bond market. We also rule out that the effects are due to the implementation of Basel III in conjunction with CCAR requirements. We believe these results yield important insights into the dynamic nature of liquidity provision in the corporate bond market.

These results also can be helpful in the current debates regarding revising or repealing the Volcker Rule. As evidenced by the regulators' attempts to exempt legitimate market making from Volcker restrictions, the original intent of the Volcker Rule was not to reduce bond market liquidity. Yet, the rules defining this exemption are cumbersome, and their implementation unwieldy, with the result that bank dealers have pulled back from corporate bond market making in stress periods post-Volcker. This could suggest, as Duffie (2012, p. 7) conjectured, that “an attempt to separate legitimate and acceptable market-making from speculative and risky market-making is not productive.” At a minimum, revising the criteria defining proscribed from permissible market making seems advisable. Alternatively, perhaps like government or municipal bond trading, corporate bond trading should be exempted from the Volcker Rule. Another possible change would be to exempt bond trading around certain events from reported metrics.³¹

Should the Volcker Rule be repealed? Our results are not sufficient to address this question. The Volcker Rule prohibits or limits banks from a range of proprietary activities such as investing in hedge funds or private equity, and the effects are simply not well understood. More careful analysis of the effects, both intended and unintended, of the Volcker Rule would seem to be in order.

Appendix

Below, we reproduce a list from Oliver Wyman and SIFMA (2011) of 21 major securities dealers and whether they are subject to the Volcker Rule. Dealers affected by the Volcker Rule are prohibited from participating in proprietary trading, but have a market-making exception. Non-affected dealers are not subject to bans on proprietary trading or market making.

Dealers affected by Volcker Rule	Dealers not affected by Volcker Rule
Bank of Nova Scotia	Cantor Fitzgerald & Co.
Barclays Capital	Daiwa Capital Markets Americas
BMO Capital Markets	Jefferies & Company
BNP Paribas Securities	Nomura Securities International
Citigroup Global Capital Markets	
Credit Suisse Securities (USA)	
Deutsche Bank Securities	
Goldman, Sachs & Co.	
HSBC Securities (USA)	
J.P. Morgan Securities	
Merrill Lynch, Pierce, Fenner & Smith	
Mizuho Securities USA	
Morgan Stanley & Co.	
RBC Capital Markets	
RBS Securities	
SG Americas Securities	
UBS Securities	

References

- Acharya, V., Richardson, M., 2009. Is the Insurance Industry Systemically Risky?. New York University, New York, Unpublished Working Paper.
- Adrian, T., Boyarchenko, N., Shachar, O., 2017. Dealer balance sheets and bond liquidity provision. *J. Monet. Econ.* 89, 92–109.
- Ambrose, B.W., Cai, N.K., Helwege, J., 2008. Forced selling of fallen angels. *J. Fixed Income* 18, 72–85.
- Ambrose, B.W., Cai, N.K., Helwege, J., 2012. Fallen angels and price pressure. *J. Fixed Income* 3, 74–86.
- Amihud, Y., 2002. Illiquidity and stock returns: cross section and time series effects. *J. Financ. Mark.* 5, 31–56.
- Anderson, M., Stulz, R.M., 2017. Is Post-Crisis Bond Liquidity Lower?. Ohio State University, Columbus, OH, Unpublished Working Paper.
- Bao, J., Edmans, A., 2011. Do investment banks matter for m&a returns? *Rev. Financ. Stud.* 24, 2286–2315.
- Bao, J., Hou, K., 2017. De facto seniority, credit risk, and corporate bond prices. *Rev. Financ. Stud.* 30, 4038–4080.
- Bao, J., Pan, J., Wang, J., 2011. The illiquidity of corporate bonds. *J. Financ.* 66, 911–946.
- Barr, M.S., Jackson, H.E., Tahyar, M.E., 2016. Financial Regulation: Law and Policy. Foundation Press, St. Paul, MN.
- Bessembinder, H., Jacobsen, S., Maxwell, W., Venkataraman, K., 2016. Capital Commitment and Illiquidity in Corporate Bonds. Arizona State University and Southern Methodist University, Tempe, AZ, and Dallas, TX, Unpublished Working Paper.
- Bessembinder, H., Kahle, K.M., Maxwell, W.F., Xu, D., 2009. Measuring abnormal bond performance. *Rev. Financ. Stud.* 22, 4219–4258.
- Brunnermeier, M.K., Nagel, S., Pedersen, L.H., 2009. Carry trades and currency crashes. *NBER Macroecon. Ann.* 23, 313–347.
- Brunnermeier, M.K., Pedersen, L.H., 2009. Market liquidity and funding liquidity. *Rev. Financ. Stud.* 22, 2201–2238.
- Cameron, A.C., Gelbach, J.B., Miller, D.L., 2011. Robust inference with multiway clustering. *J. Bus. Econ. Stat.* 29, 238–249.
- Campbell, D., Natarajan, S., 2016. Goldman Said to Prepare Volcker Defense for \$250 Million Trader. Bloomberg, November 29.
- Chernobai, A., Ozdagli, A., Wang, J., 2016. Business Complexity and Risk Management: Evidence from Operational Risk Events in US Bank Holding Companies. Syracuse University, Martin J. Whitman School of Management and Federal Reserve Bank of Boston, Syracuse, NY and Boston, MA, Unpublished Working Paper.
- Choi, J., Huh, Y., 2017. Customer Liquidity Provision: Implications for Corporate Bond Transaction Costs. University of Illinois and Board of Governors of the Federal Reserve System, Urbana-Champaign, IL, and Washington, DC, Unpublished working Paper.

³¹ We thank the legal scholars at the Wharton Conference on Financial Regulation and the Rule of Law for this suggestion.

- Deutsche Bank Research, 2016. Surviving Less Liquid Markets. Deutsche Bank Research. March 17.
- Dick-Nielsen, J., Feldhutter, P., Lando, D., 2012. Corporate bond liquidity before and after the onset of the subprime crisis. *J. Financ. Econ.* 103, 471–492.
- Dick-Nielsen, J., Rossi, M., 2016. The Cost of Immediacy for Corporate Bonds. Copenhagen Business School and Texas A&M University, Copenhagen, Denmark, and College Station, TX, Unpublished Working Paper.
- DiPietro, B., Trump victory portends major regulatory makeover. *Wall Street J.* November 10, 2016.
- Duffie, D., 2012. Market Making Under the Proposed Volcker Rule. Stanford University, Stanford, CA, Unpublished Working Paper.
- Ellul, A., Jotikasthira, C., Lundblad, C.T., 2011. Regulatory pressure and fire sales in the corporate bond market. *J. Financ. Econ.* 101, 596–620.
- Eom, Y.H., Helwege, J., Huang, J.Z., 2004. Structural models of corporate bond pricing: an empirical analysis. *Rev. Financ. Stud.* 17, 499–544.
- Federal Register, 2011. Prohibitions and Restrictions on Proprietary Trading and Certain Interests in, and Relationships with, Hedge Funds and Private Equity Funds. 76, No. 215, November 7. US Government Printing Office, Washington, DC.
- Federal Register, 2014. Prohibitions and Restrictions on Proprietary Trading and Certain Interests in, and Relationships with, Hedge Funds and Private Equity Funds. 79, No. 21, January 31. US Government Printing Office, Washington, DC.
- Federal Reserve Board, 2016. Volcker Rule: Frequently Asked Questions. <http://www.federalreserve.gov/bankinfo/volcker-rule/faq.htm>.
- Financial Stability Oversight Council, 2011. Study and Recommendations on Prohibitions on Proprietary Trading and Certain Relationships with Hedge Funds and Private Equity Funds. US Government Printing Office, Washington, DC, January.
- Finkle, V. A Dodd-Frank Rollback Bill Clears a House Committee, 2016 *New York Times* September 13.
- Friedwald, N., Jankowitsch, R., Subrahmanyam, M., 2012. Illiquidity or credit deterioration: a study of liquidity in the US corporate bond market during financial crises. *J. Financ. Econ.* 105, 18–36.
- Frydman, C., Hilt, E., 2017. Investment banks as corporate monitors in the early 20th century united states. *Am. Econ. Rev.* 107, 1938–1970.
- Goldstein, M.A., Hotchkiss, E.S., 2017. Providing Liquidity in an Illiquid Market: Dealer Behavior in US Corporate Bonds. Babson College and Boston College, Wellesley, MA, and Chestnut Hill, MA, Unpublished Working Paper.
- Hamilton, J., 2017. Deutsche Bank is First Bank Busted for Breaking Volcker Rule. Bloomberg. April 20.
- Han, S., Zhou, X., 2014. Informed bond trading, corporate yield spreads, and corporate default prediction. *Manag. Sci.* 60, 675–694.
- Harris, L., 2015. Transactions Costs, Trade Throughs, And riskless principal Trading in Corporate Bond Markets. University of Southern California Marshall School of Business, Los Angeles, CA, Unpublished Working Paper.
- Hu, G.X., Pan, J., Wang, J., 2013. Noise as information for illiquidity. *J. Financ.* 68, 2341–2382.
- Huang, J.Z., Huang, M., 2012. How much of the corporate-treasury yield spread is due to credit risk? *Rev. Asset Pricing Stud.* 2, 153–202.
- Huang, J.Z., Sun, Z., Yao, T., Yu, T., 2016. Liquidity Premium in the Eye of the Beholder: An Analysis of the Clientele Effect in the Corporate Bond Market. Pennsylvania State University, University of Massachusetts Dartmouth, University of Iowa Henry B. Tippie College of Business, and University of Cincinnati, University Park, PA, Dartmouth, MA, Iowa City, IA, and Cincinnati, OH, Unpublished Working Paper.
- Oliver, Wyman, SIFMA, 2011. The Volcker Rule Restrictions on Proprietary Trading: Implications for the US Corporate Bond Market. Securities Industry and Financial Markets Association, New York. December.
- Petersen, M.A., 2008. Estimating standard errors in finance panel data sets: comparing approaches. *Rev. Financ. Stud.* 22, 435–480.
- Powell, J.H., 2016. Trends in fixed-income markets. In: Testimony before US Senate. Committee on Banking, Housing, and Urban Affairs, Subcommittee on Securities, Insurance, and Investment and Subcommittee on Economic Policy, Washington, DC., April 14.
- Ronen, T., Zhou, X., 2013. Trade and information in the corporate bond market. *J. Financ. Mark.* 16, 61–103.
- Schultz, P., 2017. Inventory Management by Corporate Bond Dealers. University of Notre Dame, Notre Dame, IN, Unpublished Working Paper.
- Svensson, L., 1994. Estimating and Interpreting Forward Interest Rates: Sweden 1992–1994. National Bureau of Economic Research, Cambridge, MA, Unpublished Working Paper.
- Tracy, R., Carney, J., 2016. How to kill the Volcker rule? Don't enforce it. *Wall Street J.* November 28.
- Trebbi, F., Xiao, K., 2015. Regulation and Market Liquidity. University of British Columbia and University of British Columbia Sauder School of Business, Vancouver, British Columbia, Canada, Unpublished Working Paper.